

Mark schemes

Q1.

(a) $y(3.2) = y(3) + 0.2\sqrt{2 \times 3 + 5}$

M1

$$= 5 + 0.2 \times \sqrt{11}$$

A1

$$= 5.66332... = 5.6633 \text{ to 4dp}$$

Condone > 4dp

A1

3

(b) $y(3.4) = y(3) + 2(0.2) \{f[3.2, y(3.2)]\}$

M1

$$.... = 5 + 2(0.2)\sqrt{2 \times 3.2 + 5.6633...}$$

Ft on cand's answer to (a)

A1F

$$(= 5 + (0.4)\sqrt{12.0633...})$$

$$= 6.389 \text{ to 3dp}$$

CAO Must be 6.389

A1

3

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[6]

Q2.

$$k_1 = 0.2 \times (2 - 1)\sqrt{2+1} \quad (= 0.2\sqrt{3})$$

PI. May be seen within given formula.

$$= 0.346(410...) (= *)$$

Accept 3dp or better as evidence of the M1 line.

M1

$$k_2 = 0.2 \times f(2.2, 1 + *...) = 0.2 \times (2.2 - 1.346...) \sqrt{2.2+1.346...}$$

$$0.2 \times (2.2 - 1 - c's k_1) \sqrt{(2.2+1+c's k_1)}$$

PI May be seen within given formula.

M1

$$.... = 0.321(4946....)$$

A1

$$y(2.2) = y(2) + \frac{1}{2}[k_1 + k_2] = 1 + 0.5 \times [0.3464... + 0.3214...]$$

Dep on previous two Ms but ft on c's numerical values
for k_1 and k_2 following evaluation of these.

m1

$$= 1 + 0.5 \times 0.667904....$$

$$(= 1.33395...) = 1.334 \text{ to 3dp}$$

CAO Must be 1.334

SC Any consistent use of a MR / MC of printed $f(x,y)$
expression in applying IEF, mark as SC2 for a correct ft
final 3dp value otherwise SC0.

A1

[5]

Q3.

$$(a) \quad y(1.1) = y(1) + 0.1 \left[\frac{2-1}{4+1} \right]$$

M1A1

$$= 2 + 0.02 = 2.02$$

A1

3

$$(b) \quad y(1.2) = y(1) + 2(0.1)\{f[1.1, y(1.1)]\}$$

M1

$$= 2 + 2(0.1) \left[\frac{2.02-1.1}{2.02^2+1.1} \right]$$

ft on c's answer to (a)

A1F

$$= 2.035518... = 2.036 \text{ to 3dp}$$

CAO Must be 2.036

A1

3

[6]

Q4.

$$k_1 = 0.25 \times (\sqrt{2 \times 2} + \sqrt{9}) \quad (= 1.25)$$

PI. May see within given formula

M1

$$k_2 = 0.25 f(2.25, 9 + 1.25)$$

$$k_2 = 0.25 \times (\sqrt{2 \times 2.25} + \sqrt{9 + 1.25})$$

*Either $k_2 = 0.25 f(2.25, 10.25)$ stated / used
or*

$$k_2 = 0.25 \times (\sqrt{2 \times 2.25} + \sqrt{9 + \text{c's } k_1})$$

M1

$$k_2 = 1.33 (072\dots)$$

PI. May see within given formula

$$k_2 = 1.33 (072\dots) \text{ 2 dp or better}$$

PI by later work

A1

$$y(2.25) = y(2) + \frac{1}{2} [k_1 + k_2]$$

$$= 9 + 0.5 [1.25 + 1.33 (072\dots)]$$

$$= 9 + 0.5 \times 2.58 (072\dots)$$

*Dep on previous two Ms and $y(2) = 9$ and numerical
values for k 's*

m1

$$y(2.25) = 10.29036\dots = 10.29 \text{ (to 2 dp)}$$

CAO Must be 10.29

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A1

[5]

Q5.

$$k_1 = 0.1 \times (3 + \sqrt{4}) \quad (=0.5)$$

M1

$$k_2 = 0.1 f(3.1, 4.5)$$

M1

$$k_2 = 0.1 \times (3.1 + \sqrt{4.5}) = 0.522132$$

PI accept 3dp or better

A1

$$y(3.1) = y(3) + \frac{1}{2}[k_1 + k_2]$$

$$= 4 + 0.5 \times 1.022132....$$

Dep on previous two Ms and numerical values for k's

m1

$$y(3.1) = 4.511$$

Must be 4.511

A1

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Q6.

$$k_1 = 0.2 \times [2 + \ln(1 + 1)]$$

PI. May be seen within given formula

M1

$$= 0.5386(29...) \quad (= *)$$

Accept 3sf rounded or truncated or better as evidence of the M1 line

$$k_2 = 0.2 \times f(2.2, 1 + \dots)$$

$$\dots = 0.2 \times [2.2 + \ln(1 + 1.5386...)]$$

$$0.2 \times [2.2 + \ln(1 + 1 + c's k_1)].$$

PI May be seen within given formula

M1

$$\dots = 0.6263(248....)$$

4dp or better. PI by later work

A1

$$y(2.2) = y(2) + \frac{1}{2}[k_1 + k_2]$$

$$= 1 + 0.5 \times [0.5386... + 0.6263...]$$

Dep on previous two Ms but ft on c's numerical values (or numerical expressions) for k's following evaluation of these.

m1

$$= 1 + 0.5 \times 1.16495....$$

$$(= 1.582477...) = 1.5825 \text{ to 4dp}$$

CAO Must be 1.5825

SC For those scoring M1M0 who have
 $k_2 = 0.5261(78..)$, and final answer 1.5324
(ie 4 dp) for $y(2.2)$
award a total of 2 marks [M1B1]

A1

[5]

Q7.

(a) $y_1 = 2 + 0.1 \times [3 \ln(2 \times 3 + 2)] = 2 + 0.3 \ln 8$
 $= 2.6238(3...)$

M1A1

$y(3.1) = 2.6238$ (to 4 dp)

Condone greater accuracy

A1

3

(b) $k_1 = 0.1 \times 3 \ln 8 = 0.6238(32...)$

PI ft from (a), 4 dp or better

B1F

$k_2 = 0.1 \times f(3.1, 2.6238(32...))$

M1

$\dots = 0.1 \times 3.1 \times \ln 8.8238(32..)$
 $[= 0.6750(1...)]$

PI; ft on $0.1 \times 3.1 \times \ln[6.2 + \text{answer}(a)]$

A1F

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$y(3.1) = 2 + \frac{1}{2} [0.6238(3..) + 0.6750(1..)]$

m1

$= 2.6494(2...) = 2.6494$ to 4 dp

CAO Must be 2.6494

A1

5

[8]

Q8.

(a) $y(1.1) = y(1) + 0.1[1 + 3 + \sin 1]$

M1A1

$= 1 + 0.1 \times 4.84147. = 1.4841(47..) = 1.4841$ to 4 dp

Condone > 4 dp

A1

3

(b) $y(1.2) = y(1) + 2(0.1)\{f[1.1, y(1.1)]\}$

M1

$$\dots = 1 + 2(0.1)\{1.1 + 3 + \sin[1.4841(47\dots)]\}$$

Ft on cand's answer to (a)

A1F

$$= 2.019 \text{ to 3 dp}$$

CAO Must be 2.019

Note: If using degrees max mark is 4/6 ie M1A1A0; M1A1FA0

A1

3

[6]

Q9.

(a) $y_1 = 3 + 0.2 \times \left[\frac{1^2 + 3^2}{1 + 3} \right]$

M1A1

$$= 3.5$$

A1

3

(b) $k_1 = 0.2 \times 2.5 = 0.5$
PI ft from (a)

B1Ft

$$k_2 = 0.2 \times f(1.2, 3.5)$$

ft on (a)

M1

$$\dots = 0.2 \times \frac{1.2^2 + 3.5^2}{1.2 + 3.5} = 0.5825(53\dots)$$

PI condone 3 dp

A1Ft

$$y(1.2) = y(1) + \frac{1}{2}[0.5 + 0.5825(53\dots)]$$

m1

$$= 3.54127 \dots = 3.5413 \text{ to 4dp}$$

ft one slip

If answer not to 4 dp withhold this mark

A1Ft

5

[8]

Q10.

(a) $y(3.1) = y(3) + 0.1 \sqrt{3^2 + 2 + 1}$

M1A1

$$= 2 + 0.1 \times \sqrt{12} = 2.3464 \text{ (10..)} \\ = 2.3464$$

Condone > 4 dp if correct

A1

3

(b) $y(3.2) = y(3) + 2(0.1)[f(3.1, y(3.1))]$

M1

$$\dots = 2 + 2(0.1) \left[\sqrt{(3.1)^2 + 2.3464 + 1} \right]$$

ft on candidate's answer to (a)

A1F

$$\dots = 2 + 0.2 \times 3.599499\dots = 2.719(89\dots) \\ = 2.720$$

CAO Must be 2.720

A1

3

[6]

Q11.

(a) $y(2.1) = y(2) + 0.1[2^2 - 1^2]$

M1A1

$$= 1 + 0.1 \times 3 = 1.3$$

A1

3

(b) $y(2.2) = y(2) + 2(0.1)[f(2.1, y(2.1))]$

M1

$$.... = 1 + 2(0.1)[2.1^2 - 1.3^2]$$

Ft on cand's answer to (a)

A1ft

$$.... = 1 + 0.2 \times 2.72 = 1.544$$

CAO

A1

3

[6]

Q12.

$$k_1 = 0.1 \times \ln(2 + 3)$$

M1

$$= 0.1609(4379...) \quad (= *)$$

PI

A1

$$k_2 = 0.1 \times f(2.1, 3 + * ...)$$

$$... = 0.1 \times \ln(2.1 + 3.16094...)$$

M1

$$.... = 0.1660(31...)$$

PI

A1

$$y(2.1) = y(2) + \frac{1}{2} [k_1 + k_2]$$

$$= 3 + 0.5 \times 0.3269748...$$

*Dep on previous two Ms and numerical
values for k's*

m1

$$= 3.163487... = 3.1635 \text{ to 4dp}$$

Must be 3.1635

A1

[6]

Q13.

$$(a) \quad y(1.05) = 0.6 + 0.05 \times [\ln(1 + 1 + 0.6)]$$

M1A1

$$= 0.6477 (7557..) = 0.6478 \text{ to 4dp}$$

Condone > 4 dp

A1

3

(b) $k_1 = 0.05 \times \ln(1 + 1 + 0.6) = 0.0477 \text{ (75...)}$
PI

M1

ft candidate's evaluation in (a)

A1F

$$k_2 = 0.05 \times f(1.05, 0.6477...)$$

$$... = 0.05 \times \ln(1 + 1.05^2 + 0.6477...)$$

M1

$$... = 0.0505(85 ...)$$

PI

A1F

$$y(1.05) = y(1) + \frac{1}{2}[k_1 + k_2]$$

Dep on previous two Ms and numerical values for k's

m1

$$= 0.6 + 0.5 \times 0.09836...$$

$$= 0.6492 \text{ to 4dp}$$

Must be 4 dp... ft one slip

A1F

6

[9]

Q14.

(a) $y_1 = 2 + 0.1 \times \sqrt{1^2 + 2^2 + 3}$

M1

$$y(1.1) = 2 + 0.1 \times \sqrt{8}$$

A1

$$y(1.1) = 2.28284... = 2.2828 \text{ to 4dp}$$

A1

3

(b) $k_1 = 0.1 \times \sqrt{8} = 0.2828$

PI

M1
A1ft

$$k_2 = 0.1 \times f(1.1, 2.2828\dots)$$

M1

$$= 0.1 \times \sqrt{9.42137\dots} = 0.3069(425\dots)$$

PI

A1

$$y(1.1) = y(1) + \frac{1}{2} [0.28284\dots + 0.30694\dots]$$

m1

$$2.29489\dots = 2.2949 \text{ to 4dp}$$

A1

6

[9]

Q15.

(a) (i) $y(1.1) = y(1) + 0.1[1 \ln 1 + 1/1]$

M1A1

$$= 1 + 0.1 = 1.1$$

A1

3

(ii) $y(1.2) = y(1) + 2(0.1)[f(1.1, y(1.1))]$

M1A1

$$\dots = 1 + 2(0.1)[1.1 \ln 1.1 + (1.1/1.1)]$$

On answer to (a)(i)

A1 ✓

$$\dots = 1 + 0.2 \times 1.104841198 \dots$$

$$\dots = 1.22096824 = 1.221 \text{ to 3 dp}$$

CAO

A1

4

(b) (i) IF is $e^{\int -\frac{1}{x} dx}$

Condone $e^{\int \frac{1}{x} dx}$ for M mark

M1

$$= e^{-\ln x}$$

A1

$$= e^{\ln x^{-1}} = x^{-1} = \frac{1}{x}$$

AG (be convinced)

(b)(i) Solutions using the printed answer must be convincing before any marks are awarded

A1

3

(ii) $\frac{d}{dx} \left(\frac{y}{x} \right) = \ln x$

M1A1

$$\frac{y}{x} = \int \ln x \, dx = x \ln x - \int x \left(\frac{1}{x} \right) dx$$

Integration by parts for $x^k \ln x$

M1

$$\frac{y}{x} = x \ln x - x + c$$

Condone missing c.

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A1

$$y(1) = 1 \Rightarrow 1 = \ln 1 - 1 + c$$

Dependent on at least one of the two previous M marks

m1

$$\Rightarrow c = 2 \Rightarrow y = x^2 \ln x - x^2 + 2x$$

$$\text{OE eg } \frac{y}{x} = x \ln x - x + 2$$

A1

6

(iii) $y(1.2) = 1.222543... = 1.223$ to 3dp

B1

1

Q16.

$$(a) \quad y_1 = 2 + 0.1 \times \left[\frac{1^2 + 2^2}{1 \times 2} \right]$$

M1A1

$$= 2 + 0.1 \times 2.5 = 2.25$$

A1

3

$$(b) \quad k_1 = 0.1 \times 2.5 = 0.25$$

M1

PI ft from (a)

A1ft

$$k_2 = 0.1 \times f(1.1, 2.25)$$

M1

$$\dots = 0.1 \times 2.53434\dots = 0.2534(34\dots)$$

PI

A1ft

$$y(1.1) = y(1) + \frac{1}{2} [0.25 + 0.253434\dots]$$

m1

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$$= 2.2517 \text{ to 4dp}$$

If answer not to 4dp withhold this mark

A1ft

[9]

Examiner reports

Q1.

Numerical solutions of first order differential equations are a good source of marks for all students. Once again, this topic was the best answered on the paper. The most common error involved an incorrect evaluation of $f[3.2, y(3.2)]$. Students lost fewer marks if the correct substitution into the formula had been shown before the incorrect evaluation. A slightly higher proportion of students than last year failed to give their final answer to the required degree of accuracy.

Q2.

This question on numerical solutions of first order differential equations again proved to be a good source of marks for many students. Almost all students found the correct value for k_1 but a significant minority did not get the correct value for k_2 . Such students generally either scored a total of 1 mark or 3 marks for the question. The greater award was normally given to those who showed a correct substitution into the formula for k_2 , for example $0.2 \times (2.2 - 1 - 0.346\dots) \sqrt{2.2+1+0.346\dots}$, before stating and using the wrong value for k_2 in the improved Euler formula. Those who just stated and used the wrong value of k_2 did not score these extra two marks. Almost all students gave their final answer to the specified degree of accuracy.

Q3.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates. Once again, this topic was the best answered on the paper. The most common error involved an incorrect evaluation of $f[1.1, y(1.1)]$. Almost all candidates gave their final answer to the required degree of accuracy.

Q4.

This question on numerical solutions of first order differential equations proved a welcome opener again, with the majority of students scoring full marks. Almost all students gave their final answer to the specified degree of accuracy. It is advisable for students to show full working with clear substitutions into relevant formulae before carrying out the evaluations.

For example, a number of students did not show $k_2 = 0.25 \times (\sqrt{2 \times 2.25} + \sqrt{9 + 1.25})$ and instead wrote a similar statement but not with both 4.5 and 10.25 under the square root signs. Without evidence of a correct method, such solutions were heavily penalised.

Q5.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates, and it was the best answered topic on the paper. However, a few candidates who had an incorrect value for k_1 and just gave a table of values without showing any methods gained no credit. Almost all candidates gave their final answer to the required degree of accuracy.

Q6.

Numerical solutions of first order differential equations continue to be a good source of marks with most candidates showing full working with clear substitutions into relevant

formulae. Unfortunately, more candidates than usual made arithmetical errors in calculating the value of k_2 . Very few candidates failed to give their final answer to the specified degree of accuracy.

Q7.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates and this was the best answered question on the paper. Very few candidates mixed up the x and y values in applying the given formulae. Almost all candidates gave their final answers to the required degree of accuracy.

Q8.

This was the best answered question on the paper. Numerical solutions of first order differential equations continue to be a good source of marks for all candidates. The most common loss of marks was due to calculators being set in degree mode. It is worth recording that slightly more candidates than in recent series have slipped back into not showing the necessary working. Without such working, wrong answers cannot be awarded credit.

Q9.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates. This was the best answered question on the paper and it was pleasing to see full working with clear substitutions into relevant formulae. The vast majority of candidates obtained the correct answers, and errors were generally limited to incorrect evaluations rather than any lack of understanding of the methods or notation involved.

Q10.

Numerical solutions of first order differential equations continue to be a good source of marks for all candidates. Although it was the best answered question on the paper, more candidates than usual mixed up the x and y values in applying the given formulae, or incorrectly used $f(3, 2)$ instead of $f(3.1, y(3.1))$ in the given formula in part (b). There were very few candidates who lost the final accuracy mark for failing to give their answer correct to three decimal places.

Q11.

Numerical solutions of first order differential equations continues to be a good source of marks for all candidates. This was the best answered question on the paper. Although almost all candidates obtained the correct answer to part (a), some less able candidates showed a lack of understanding of the notation used in the given formula in part (b).

Q12.

The majority of candidates were able to correctly use the given improved Euler formula to find the approximate value of $y(2.1)$ to four decimal places. Otherwise, the most common error was to use $y_r + h$ instead of $y_r + k_1$ in writing k_2 as $0.1 \ln(2.1 + 3.1)$. There was a minority of candidates who failed to gain marks because they gave the wrong answer for $y(2.1)$ and showed no method in their working, instead just giving a table of incorrect values.

Q13.

This question proved to be a good source of marks to all those candidates who showed sufficient working. There was some evidence of misreads, the most common one being “ $\ln(1 + x^2 + y^2)$ ” for $f(x, y)$; for those candidates who explicitly showed this misread, the mark loss was minimal.

Q14.

This question which tested numerical solutions of differential equations was answered well with a large proportion of candidates gaining full marks. The most common error was the use of 2.1 instead of 2.2828... in finding k_2 . Although there were again candidates who failed to show working (by writing down the correct formula and then substituting in the correct values), the proportion doing so was significantly less than last year. The vast majority of candidates gave their final answers to the required degree of accuracy but more should be encouraged to show intermediate answers to more than the stated degree of accuracy.

Q15.

This question was a good source of marks for many candidates. The methods in part (a) were well understood and most candidates worked accurately to obtain the correct approximations. In part (b)(i) a high proportion of the candidates obtained the printed answer but some failed to show sufficient stages in their working. In part (b)(ii), weaker candidates could not integrate $\ln x$; such candidates frequently gave the answer as $1/x$. Those who inserted the constant of integration were generally able to apply the given boundary condition. Surprisingly, not all candidates who obtained the correct solution for part (b)(ii) were able to give the correct value for y in part (b)(iii).

Q16.

This question which tested numerical solutions of differential equations was the best answered question on the paper. The most common error was the use of 2.1 instead of 2.25 in finding k_2 . A significant number of candidates presented their values in the form of a well-labelled table but some included incorrect values with no indication where the error had occurred. As a result the method marks could not be awarded. Very few candidates failed to give their final answer to the required degree of accuracy.